

SMART HOME AUTOMATION USING BLYNK APP

Team Members:

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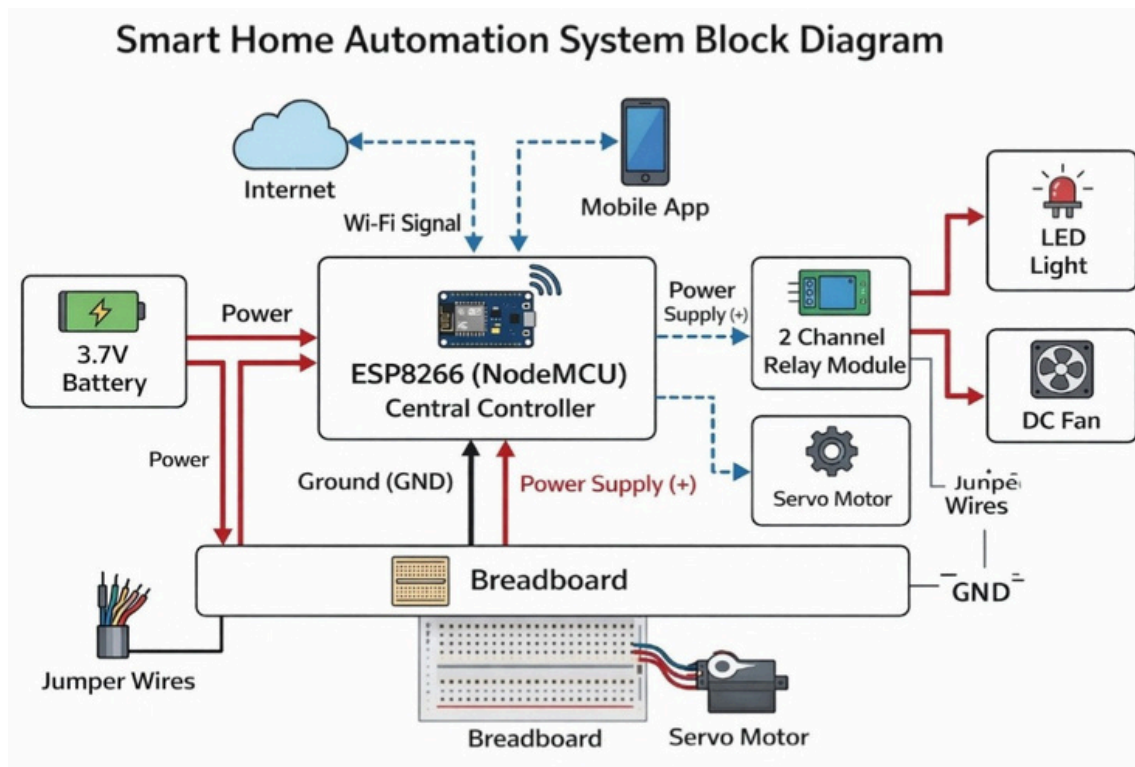
Abstract:

Smart Home Automation using IoT is a system that allows users to control home appliances remotely using the internet. This project uses ESP8266 to connect devices like lights and fans to a mobile application. It improves convenience, saves energy, and enhances home security. Users can monitor and control appliances from anywhere. This system makes everyday life easier and smarter.

Apparatus:

- ESP8266 (NodeMCU)
- 2 Channel Relay Module
- Jumper Wires
- Breadboard
- LED
- Fan
- 3.7V Battery

Block Diagram:



Program:

```
/* 1. Project Setup Information */
#define BLYNK_TEMPLATE_ID "TMPL3Jvg8BWa7"
#define BLYNK_TEMPLATE_NAME "iot"
#define BLYNK_AUTH_TOKEN "2L9FauWm8gCpf_tyBDDroUXKLIMkCpQN"

/* 2. Libraries */
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

/* 3. WiFi Credentials */
char ssid[] = "vivo 1606";
char pass[] = "9884836915";

/* 4. Hardware Pin Definitions */
const int ledPin = D1; // Connect your LED Relay to D1
const int fanPin = D2; // Connect your Fan Relay to D2

// -----
// LED CONTROL (Virtual Pin V0)
// -----
BLYNK_WRITE(V0) {
  int state = param.asInt(); // Get value from Blynk App (0 or 1)

  if (state == 1) {
    digitalWrite(ledPin, LOW); // Turn LED Relay ON
    Serial.println("LED is now ON");
  } else {
    digitalWrite(ledPin, HIGH); // Turn LED Relay OFF
    Serial.println("LED is now OFF");
  }
}

// -----
// FAN CONTROL (Virtual Pin V1)
// -----
BLYNK_WRITE(V1) {
  int state = param.asInt(); // Get value from Blynk App (0 or 1)
```

```

if (state == 1) {
  digitalWrite(fanPin, LOW); // Turn Fan Relay ON
  Serial.println("Fan is now ON");
} else {
  digitalWrite(fanPin, HIGH); // Turn Fan Relay OFF
  Serial.println("Fan is now OFF");
}
}

// -----
// INITIAL SETUP
// -----

void setup() {
  Serial.begin(115200);

  // Set pins as OUTPUT
  pinMode(ledPin, OUTPUT);
  pinMode(fanPin, OUTPUT);

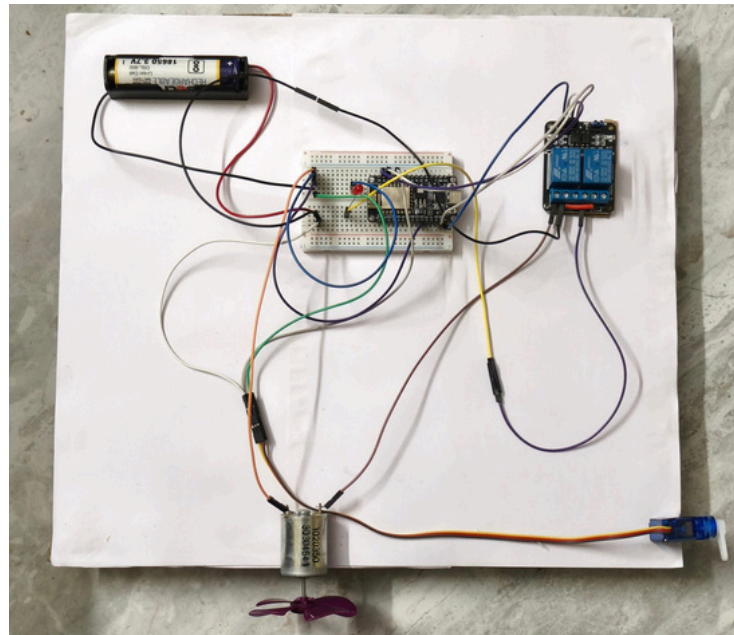
  // Keep them OFF initially (HIGH signal for most relays)
  digitalWrite(ledPin, HIGH);
  digitalWrite(fanPin, HIGH);

  // Connect to WiFi and Blynk
  Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);
  Serial.println("System Ready!");
}

void loop() {
  Blynk.run(); // Keeps the connection to the Blynk app alive
}

```

Output:



Conclusion:

Smart Home Automation using IoT is an efficient system that provides comfort, security, and energy saving. It reduces manual work and allows users to control appliances easily from anywhere. This project shows how IoT can improve modern living. It is cost-effective and can be expanded with more features in the future.